Table R1: **Table 2 of the submission with the refined rule ($\delta = 0.3$) as the default.** Counterfactual Video Foley Generation on 11,890 VGGSound-Counterfactual triplets. Panel (b) uses comparable NFE, with doubled steps for Default CFG and negative-source CFG and the published transition for CounterFlow. Bold and underline mark the best and second-best values *across all rows in (a)* and *within each backbone in (b)*. Blue marks the new values of the refined rule: the **TAP-Foley** rows use the margin rule with $\delta = 0.3$. The rows marked "$\alpha = 1.2$, submitted" are the submitted scaled projection, kept for reference with their submitted values (black) and not ranked. Baseline rows, including their bold and underline marks, are exactly as submitted. †FAD is not ranked.

Method		Sound Identity Replacement					Temporal Alignment	Distribution /Quality		
		CLAP-T ↑	Clf. acc. ↑	FLAM-T ↑	ΔFLAM ↑	(+) Ratio ↑	DeSync ↓	IS ↑	PQ ↑	FAD†
(a) Comparison with existing VT2A systems										
Ours										
TAP-Foley	MMAudio	**0.299**	**0.901**	**0.960**	0.425	**0.916**	0.455	**21.16**	6.21	22.8
	Omni2Sound	0.242	0.833	0.891	0.301	0.824	0.501	19.98	6.30	24.7
TAP-Foley (α=1.2, submitted)	MMAudio	0.296	0.885	0.959	0.384	0.900	0.427	20.64	6.21	21.6
	Omni2Sound	0.241	0.829	0.887	0.299	0.821	0.510	19.83	6.31	24.5
Baseline										
ControlFoley		0.229	0.601	0.864	0.089	0.610	**0.316**	17.10	6.13	14.6
CAFA		0.243	0.786	0.942	0.194	0.785	0.685	12.21	6.36	22.3
FoleyCrafter		0.154	0.514	0.746	-0.004	0.478	1.244	10.74	6.06	30.4
ReWaS		0.005	0.443	0.344	-0.005	0.504	1.088	5.33	5.44	58.3
AudioX		0.187	0.475	0.840	-0.034	0.475	1.136	16.27	**6.53**	18.1
Baseline (with negative-source CFG)										
ControlFoley		0.255	**0.893**	0.910	**0.456**	**0.916**	0.754	15.66	5.89	23.6
CAFA		0.200	0.873	0.875	0.335	0.860	0.715	10.36	6.25	24.5
FoleyCrafter		0.105	0.732	0.662	0.281	0.742	1.264	8.08	6.02	40.8
ReWaS		-0.045	0.586	0.190	0.050	0.575	1.168	4.73	5.52	67.7
AudioX		0.217	0.845	0.873	0.329	0.853	1.233	14.79	6.19	25.2
(b) Comparison of inference-time guidance methods										
MMAudio										
TAP-Foley (ours)		**0.299**	0.901	**0.960**	0.425	0.916	0.455	**21.16**	**6.21**	22.8
TAP-Foley (α=1.2, submitted)		0.296	0.885	0.959	0.384	0.900	0.427	20.64	6.21	21.6
CounterFlow		0.292	**0.916**	0.949	**0.475**	**0.927**	0.649	19.75	6.16	23.7
Default CFG		0.162	0.350	0.727	-0.145	0.335	**0.323**	16.39	**6.21**	10.8
Negative-source CFG		0.212	0.758	0.824	0.280	0.778	0.703	12.93	5.93	19.2
Omni2Sound										
TAP-Foley (ours)		**0.242**	**0.833**	**0.891**	**0.301**	**0.824**	0.501	**19.98**	**6.30**	24.7
TAP-Foley (α=1.2, submitted)		0.241	0.829	0.887	0.299	0.821	0.510	19.83	6.31	24.5
CounterFlow		0.213	0.765	0.862	0.226	0.740	0.576	15.00	6.06	23.7
Default CFG		0.110	0.223	0.620	-0.267	0.195	**0.408**	18.58	6.27	14.2
Negative-source CFG		0.132	0.453	0.714	-0.029	0.438	0.642	11.35	6.05	27.1

Table R2: **Table 7 of the submission with the refined rule** ($\delta = 0.3$)**.** Onset-level evaluation on 1,280 Greatest Hits counterfactual triplets. Onset scores count only detected onsets with PE-A-Frame evidence for the target identity, so they measure identity and timing jointly (target-filtered onset metrics, appendix of the submission). Recall, precision, and F1 are the standard event-level scores over reference onsets, from a maximum one-to-one matching within a 0.3-s tolerance. Panel (a) covers the two conflict-aware systems. Panel (b) uses comparable NFE and the published transition for CounterFlow. Bold and underline mark the best and second-best values *across all rows in (a)* and *within each backbone in (b)*. Blue marks the new values of the refined rule: the **TAP-Foley** rows use the margin rule with $\delta = 0.3$. The rows marked "$\alpha = 1.2$, submitted" are the submitted scaled projection, kept for reference with their submitted values (black) and not ranked. Baseline rows, including their bold and underline marks, are exactly as submitted.

Method		Sound Identity Replacement		Onset Analysis		
		CLAP-T ↑	Clf. acc. ↑	Onset recall ↑	Onset precision ↑	Onset F1 ↑
(a) Comparison with existing VT2A systems						
Ours						
TAP-Foley	MMAudio	**0.347**	**0.997**	**0.864**	0.802	**0.797**
	Omni2Sound	0.303	0.952	0.809	0.775	0.757
TAP-Foley (α=1.2, submitted)	MMAudio	0.343	0.991	0.851	0.818	0.801
	Omni2Sound	0.297	0.948	0.803	0.780	0.758
Baseline						
ControlFoley		0.301	0.881	0.771	**0.833**	0.769
CAFA		0.305	0.956	0.775	0.802	0.756
Baseline (with negative-source CFG)						
ControlFoley		0.305	0.981	0.754	0.632	0.616
CAFA		0.276	0.976	0.767	0.722	0.704
(b) Comparison of inference-time guidance methods						
MMAudio						
TAP-Foley		**0.347**	**0.997**	**0.864**	0.802	**0.797**
TAP-Foley (α=1.2, submitted)		0.343	0.991	0.851	0.818	0.801
CounterFlow		0.305	0.981	0.705	0.673	0.631
Default CFG		0.295	0.877	0.746	**0.830**	0.751
Negative-source CFG		0.324	0.984	0.798	0.724	0.716
Omni2Sound						
TAP-Foley		**0.303**	**0.952**	**0.809**	**0.775**	**0.757**
TAP-Foley (α=1.2, submitted)		0.297	0.948	0.803	0.780	0.758
CounterFlow		0.249	0.913	0.713	0.632	0.612
Default CFG		0.174	0.541	0.495	0.574	0.506
Negative-source CFG		0.184	0.823	0.603	0.639	0.581

Table R3: **Family-stratified results on the full VGGSound-Counterfactual benchmark (refined rule, $\delta = 0.3$, matched NFE).** Each of the 11,890 pairs (video, target prompt) is assigned to one stratum per axis: the family of the source category, the family of the target prompt (six AV-JEPA families and Other), and the family relation (within-family: source and target in the same family). Per stratum: the number of source videos and pairs, the mean ΔFLAM and DeSync of **TAP-Foley**, and the four paired differences **TAP-Foley** minus Default CFG and **TAP-Foley** minus CounterFlow in ΔFLAM (positive favours **TAP-Foley**) and DeSync (negative favours **TAP-Foley**). Small brackets give 95% percentile confidence intervals from a cluster bootstrap over the source videos within the stratum (10,000 draws); **bold**: the interval excludes 0; "(small)": fewer than 30 videos. The first row of each block is the full benchmark (Table 2). No stratum shows a significant difference of opposite sign to the full benchmark: the four differences are significant with the sign of the full benchmark in 16/16/14/16 of the 16 strata on MMAudio and in 16/15/13/14 on Omni2Sound (column order), and not significant in the remaining strata. Within-family replacement gives lower ΔFLAM than cross-family replacement on both backbones.

Stratum	Videos	Pairs	**TAP-Foley** (margin, $\delta = 0.3$)		**TAP-Foley** − Default CFG		**TAP-Foley** − CounterFlow	
			ΔFLAM ↑	DeSync ↓	ΔFLAM	DeSync	ΔFLAM	DeSync
MMAudio								
All videos (full benchmark)	1,189	11,890	0.425	0.455	**+0.570** [+0.552, +0.589]	**+0.133** [+0.121, +0.145]	**-0.049** [-0.056, -0.042]	**-0.193** [-0.205, -0.181]
Source family								
Animals	189	1,890	0.483	0.519	**+0.550** [+0.507, +0.595]	**+0.167** [+0.138, +0.198]	**-0.057** [-0.073, -0.041]	**-0.214** [-0.242, -0.187]
Human voice/vocal	88	880	0.402	0.234	**+0.690** [+0.627, +0.753]	**+0.117** [+0.087, +0.150]	**-0.111** [-0.147, -0.077]	**-0.192** [-0.238, -0.149]
Instruments	449	4,490	0.323	0.364	**+0.484** [+0.454, +0.515]	**+0.101** [+0.084, +0.119]	**-0.044** [-0.053, -0.034]	**-0.169** [-0.188, -0.150]
Other	334	3,340	0.529	0.517	**+0.647** [+0.614, +0.680]	**+0.151** [+0.128, +0.173]	**-0.045** [-0.058, -0.033]	**-0.199** [-0.222, -0.177]
Sports	43	430	0.322	0.441	**+0.465** [+0.391, +0.544]	**+0.113** [+0.051, +0.176]	-0.008 [-0.048, +0.034]	**-0.254** [-0.316, -0.193]
Vehicles	65	650	0.567	0.852	**+0.769** [+0.686, +0.853]	**+0.202** [+0.150, +0.255]	-0.015 [-0.042, +0.010]	**-0.185** [-0.235, -0.137]
Water/weather (small)	21	210	0.311	0.582	**+0.482** [+0.397, +0.567]	**+0.101** [+0.022, +0.185]	**-0.091** [-0.168, -0.016]	**-0.353** [-0.463, -0.246]
Target family								
Animals	575	2,229	0.477	0.458	**+0.609** [+0.570, +0.650]	**+0.167** [+0.144, +0.190]	**-0.080** [-0.096, -0.064]	**-0.203** [-0.228, -0.179]
Human voice/vocal	166	554	0.456	0.258	**+0.780** [+0.705, +0.854]	**+0.104** [+0.071, +0.141]	**-0.063** [-0.099, -0.024]	**-0.143** [-0.191, -0.099]
Instruments	636	4,158	0.313	0.332	**+0.421** [+0.395, +0.447]	**+0.098** [+0.081, +0.115]	**-0.045** [-0.054, -0.036]	**-0.177** [-0.197, -0.158]
Other	715	3,319	0.492	0.570	**+0.638** [+0.608, +0.669]	**+0.142** [+0.121, +0.164]	**-0.030** [-0.043, -0.017]	**-0.216** [-0.237, -0.194]
Sports	258	707	0.353	0.502	**+0.580** [+0.521, +0.641]	**+0.168** [+0.128, +0.209]	**-0.050** [-0.082, -0.019]	**-0.202** [-0.248, -0.155]
Vehicles	343	617	0.633	0.678	**+0.840** [+0.776, +0.902]	**+0.166** [+0.115, +0.216]	**-0.050** [-0.081, -0.020]	**-0.197** [-0.245, -0.150]
Water/weather	121	306	0.549	0.672	**+0.643** [+0.548, +0.740]	**+0.154** [+0.085, +0.226]	**-0.063** [-0.117, -0.011]	**-0.159** [-0.230, -0.085]
Family relation								
Within-family	1,145	8,060	0.366	0.401	**+0.457** [+0.438, +0.476]	**+0.108** [+0.096, +0.121]	**-0.044** [-0.052, -0.036]	**-0.189** [-0.202, -0.175]
Cross-family	827	3,830	0.550	0.568	**+0.809** [+0.780, +0.839]	**+0.184** [+0.163, +0.205]	**-0.060** [-0.073, -0.047]	**-0.203** [-0.224, -0.182]
Omni2Sound								
All videos (full benchmark)	1,189	11,890	0.301	0.501	**+0.568** [+0.549, +0.588]	**+0.094** [+0.081, +0.107]	**+0.075** [+0.066, +0.083]	**-0.075** [-0.086, -0.064]
Source family								
Animals	189	1,890	0.387	0.593	**+0.651** [+0.601, +0.702]	**+0.117** [+0.082, +0.150]	**+0.081** [+0.058, +0.105]	**-0.102** [-0.128, -0.077]
Human voice/vocal	88	880	0.428	0.319	**+0.829** [+0.762, +0.898]	**+0.093** [+0.051, +0.139]	**+0.136** [+0.100, +0.174]	**-0.046** [-0.079, -0.015]
Instruments	449	4,490	0.233	0.406	**+0.497** [+0.465, +0.529]	**+0.093** [+0.072, +0.114]	**+0.082** [+0.071, +0.094]	**-0.032** [-0.047, -0.017]
Other	334	3,340	0.326	0.556	**+0.558** [+0.524, +0.592]	**+0.093** [+0.069, +0.117]	**+0.057** [+0.041, +0.073]	**-0.108** [-0.131, -0.085]
Sports	43	430	0.162	0.482	**+0.298** [+0.225, +0.374]	**+0.073** [+0.015, +0.130]	**+0.071** [+0.038, +0.106]	-0.027 [-0.066, +0.010]
Vehicles	65	650	0.336	0.754	**+0.684** [+0.602, +0.766]	+0.023 [-0.055, +0.100]	+0.029 [-0.013, +0.072]	**-0.151** [-0.210, -0.093]
Water/weather (small)	21	210	0.218	0.874	**+0.611** [+0.520, +0.709]	**+0.194** [+0.078, +0.320]	+0.032 [-0.033, +0.092]	**-0.205** [-0.273, -0.136]
Target family								
Animals	575	2,229	0.432	0.549	**+0.686** [+0.645, +0.729]	**+0.143** [+0.115, +0.171]	**+0.072** [+0.052, +0.091]	**-0.084** [-0.107, -0.061]
Human voice/vocal	166	554	0.432	0.319	**+0.838** [+0.758, +0.913]	**+0.056** [+0.021, +0.094]	**+0.146** [+0.097, +0.193]	**-0.051** [-0.086, -0.019]
Instruments	636	4,158	0.221	0.371	**+0.437** [+0.408, +0.467]	**+0.076** [+0.057, +0.094]	**+0.074** [+0.063, +0.085]	**-0.039** [-0.054, -0.024]
Other	715	3,319	0.261	0.610	**+0.590** [+0.558, +0.623]	**+0.088** [+0.064, +0.112]	**+0.064** [+0.046, +0.081]	**-0.116** [-0.138, -0.094]
Sports	258	707	0.257	0.472	**+0.504** [+0.442, +0.572]	**+0.083** [+0.043, +0.124]	**+0.075** [+0.049, +0.102]	-0.014 [-0.046, +0.019]
Vehicles	343	617	0.402	0.671	**+0.697** [+0.631, +0.764]	**+0.085** [+0.040, +0.130]	**+0.115** [+0.076, +0.155]	**-0.119** [-0.161, -0.078]
Water/weather	121	306	0.517	0.806	**+0.653** [+0.559, +0.753]	**+0.154** [+0.083, +0.224]	+0.018 [-0.019, +0.057]	**-0.146** [-0.210, -0.085]
Family relation								
Within-family	1,145	8,060	0.244	0.456	**+0.469** [+0.448, +0.489]	**+0.079** [+0.065, +0.093]	**+0.069** [+0.059, +0.078]	**-0.074** [-0.087, -0.062]
Cross-family	827	3,830	0.421	0.598	**+0.777** [+0.744, +0.809]	**+0.126** [+0.102, +0.150]	**+0.087** [+0.071, +0.103]	**-0.076** [-0.094, -0.057]